

RRC-Raptor-V4H

DDR

- RTX_SAMPLE

V0.0.1

Contents

i.	VERSION LIST	4
ii.	HASHTAG	5
1.	Overview.....	6
2.	查看 DDR 容量.....	7
3.	測試 DDR.....	8
4.	APPENDIX	10

LIST OF FIGURES

FIGURE 1. EXAMPLE.....	7
FIGURE 2. MEMORY STATUS BEFORE TEST	8
FIGURE 3 MEMORY STATUS DURING TEST	9
FIGURE 4 MEMORY TEST RESULT.....	9

i. **VERSION LIST**

Description	Version	Date	Author
RTX_DDR	V0.0.1	2024/06/26	Jerry

ii. HASHTAG

#V4H

#DDR

#BSP

#ubuntu

1. Overview

Target：介紹如何查看 DDR size, 以及如何使用 stressapptest 工具測試 DDR.

2. 查看 DDR 容量

用 `free` 指令可以查看 DDR 容量.如 Figure 1, 其中 7304 代表 7304mega bytes, 由此可知, DDR size 為 8G.

```
root@v4x:~# free -m
              total        used        free      shared  buff/cache   available
Mem:           7304          618         6605           19           80          6546
Swap:            0             0             0
```

Figure 1. Example

3. 測試 DDR

可以使用 stressapptest 來測試存取資料時 DDR 狀況.

以下為常用 stressapptest 參數的介紹

- s seconds : Number of seconds to run
- M mbytes : Megabytes of ram to test
- m threads : Number of memory copy threads to run
- C threads : Number of memory CPU stress threads to run.
- W : Use more CPU-stressful memory copy

所以我們如果要測 4G 的 DDR, 測 1 小時, 可以用以下指令來測試

```
stressapptest -s 3600 -M 4096 -m 4 -W
```

User 也可以根據自己需求調整參數.

下面為實際測試狀況.

1. 測試前先用 top 指令看一下現在系統 mem 使用狀況.現在只有 619 mega bytes 在使用.

```
top - 03:41:27 up 1:55, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 135 total, 1 running, 134 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.1 hi, 0.0 si, 0.0 st
MiB Mem : 7304.9 total, 6605.4 free, 619.3 used, 80.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 6546.4 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
455	root	20	0	31216	2928	2344	S	0.3	0.0	0:03.98	tcf-agent
560	root	20	0	4824	2568	2068	R	0.3	0.0	0:00.20	top
1	root	20	0	9188	6400	4852	S	0.0	0.1	0:01.27	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.01	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp

Figure 2. Memory status before Test

2. 在執行 stressapptest 後，已經變成 4722 在使用了，代表 stressapptest 正在測試這些記憶體。

```

root@v4x:~# stressapptest -s 3600 -M 4096 -m 4 -W
2024/06/26-03:48:56(UTC) Log: Commandline - stressapptest -s 3600 -M 4096 -m 4 -W
2024/06/26-03:48:56(UTC) Stats: SAT revision 1.0.9_autoconf, 64 bit binary
2024/06/26-03:48:56(UTC) Log: @ jerry-pc on Tue Jun 25 05:27:46 UTC 2024 from open source release
2024/06/26-03:48:56(UTC) Log: 1 nodes, 4 cpus.
2024/06/26-03:48:56(UTC) Log: Prefer plain malloc memory allocation.
2024/06/26-03:48:56(UTC) Log: Using mmap() allocation at 0xffffea40000000.
2024/06/26-03:48:56(UTC) Stats: Starting SAT, 4096M, 3600 seconds
2024/06/26-03:48:58(UTC) Log: region number 39 exceeds region count 1
2024/06/26-03:48:58(UTC) Log: Region mask: 0x1
2024/06/26-03:49:08(UTC) Log: Seconds remaining: 3590

jerry@jerry: ~ 98x15
top - 03:49:14 up 2:03, 2 users, load average: 0.89, 0.20, 0.06
Tasks: 136 total, 1 running, 135 sleeping, 0 stopped, 0 zombie
%Cpu(s): 99.2 us, 0.4 sy, 0.0 ni, 0.0 id, 0.0 wa, 0.3 hi, 0.1 si, 0.0 st
MiB Mem : 7304.9 total, 2501.8 free, 4722.9 used, 80.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 2442.8 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
  562 root        20   0 4838904  4.0g 2588 S 398.3 56.1 1:09.47 stressapptest
  560 root        20   0  4824    2568 2068 R   0.3   0.0  0:01.98 top
    1 root        20   0  9188    6400 4852 S   0.0   0.1  0:01.28 systemd
    2 root        20   0      0      0      0 S   0.0   0.0  0:00.01 kthreadd
    3 root        0 -20      0      0      0 I   0.0   0.0  0:00.00 rcu_gp
    4 root        0 -20      0      0      0 I   0.0   0.0  0:00.00 rcu_par_gp
    5 root        20   0      0      0      0 I   0.0   0.0  0:00.10 kworker/0:0-events
    8 root        0 -20      0      0      0 I   0.0   0.0  0:00.00 mm_percpu_wq

```

Figure 3 Memory status during test

3. 最後可由結束訊息中查看在測試過程中是否有 error 產生。

```

2024/06/26-03:55:59(UTC) Stats: Found 0 hardware incidents
2024/06/26-03:55:59(UTC) Stats: Completed: 357004.00M in 60.01s 5949.43MB/s, with 0 hardware incidents, 0 errors
2024/06/26-03:55:59(UTC) Stats: Memory Copy: 357004.00M at 5949.80MB/s
2024/06/26-03:55:59(UTC) Stats: File Copy: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Stats: Net Copy: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Stats: Data Check: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Stats: Invert Data: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Stats: Disk: 0.00M at 0.00MB/s
2024/06/26-03:55:59(UTC) Status: PASS - please verify no corrected errors
2024/06/26-03:55:59(UTC)

```

Figure 4 Memory test result

4. APPENDIX